TESTING AND QUALITY MANAGEMENT TOOLS SUPPLEMENTAL GUIDE

This document contains OUM supplemental guidance for using Oracle® Application Test Suite (OATS) Testing and Quality Management Tools in conjunction with the Oracle Unified Method (OUM). OATS is a complementary Oracle suite of tools that is used to manage software testing for a project as well as to develop test automation and performance testing, when used together these tools can be used to maximize system performance and ensure the quality and success of a project.

The Testing and Quality Management Tools Supplemental Guide contains the following components:

Testing and Quality Management Tools Overview

This component provides an overview of of some of the Testing and Quality Management Tools.

Testing and Quality Management Tools Process Guidelines

The Process Guidelines should be used in conjunction with the standard OUM process guidelines when using Testing and Quality Management Tools.

Testing and Quality Management Tools Supplemental Task Guidelines

The Supplemental Task Guidelines should be used in conjunction with the standard OUM tasks when using Testing and Quality Management Tools.

PDF View

You may also prefer to access this information through the **Testing and Quality Management Tools Supplemental Guide (PDF View)** located in the Supplemental Guides section of the Supplemental Guidance page.

Back to Top

PURPOSE

This document contains OUM supplemental guidance for using the Testing and Quality Management Tools in conjunction with OUM. The Testing and Quality

Management Tools is a complementary Oracle suite of tools that are used to manage software testing for a project as well as to develop test automation and performance testing. When used together, these tools can maximize system performance and ensure the quality and success of a project.

Back to Top

AUDIENCE

This document is intended for practitioners who will be using the Testing and Quality Management Tools in conjunction with any project using the OUM approach.

Back to Top

HOW TO USE THIS DOCUMENT

This document contains supplemental guidance for using the Testing and Quality Management Tools in conjunction with and not instead of the OUM base method guidance.

Back to Top

LIMITATIONS TO THIS DOCUMENT

This document does not describe OUM in detail – the reader must review the documentation and training materials referenced below to gain a basic understanding of each.

Back to Top

RELATED DOCUMENTATION AND TRAINING

The following related documentation and training provide additional detailed guidance related to OUM implementations and are prerequisites to this document:

- OUM Method Pack – Desktop Installable or Hosted version
- OUM Training and Accreditation Program

Back to Top

CONTRIBUTORS

Sieds Aalberts, Alex Amat, Roderick Jansen

Back to Top

ORACLE

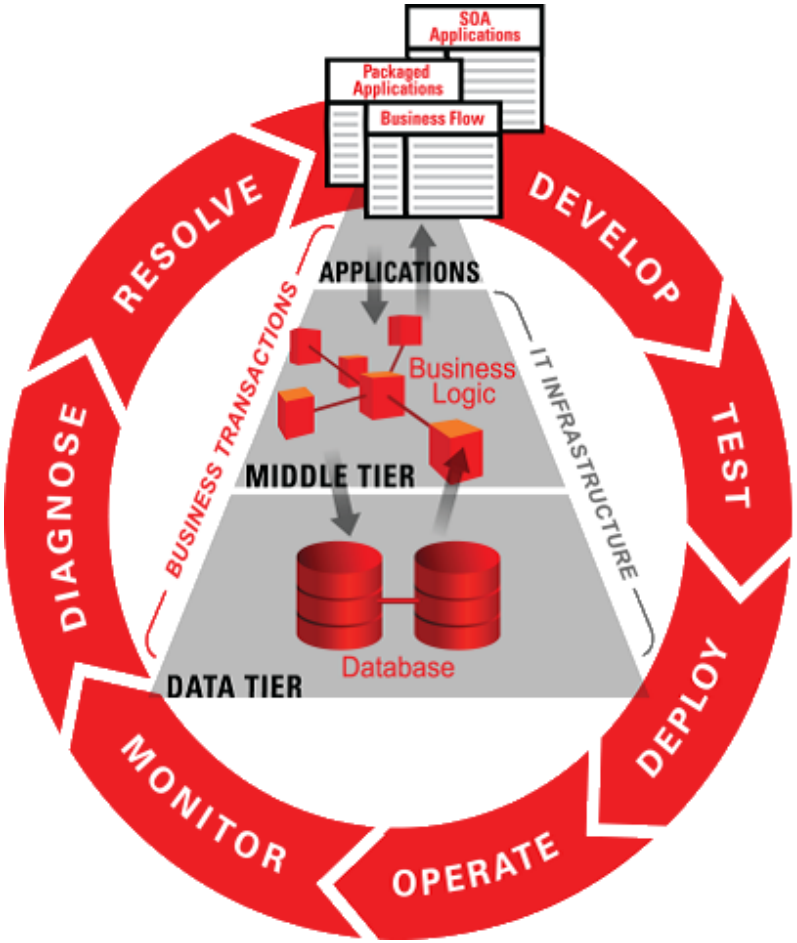
Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.

TESTING AND QUALITY MANAGEMENT TOOLS OVERVIEW

TOOL OVERVIEW

Oracle Application Testing Suite (OATS) is an open and integrated scripting platform for both load and functional testing as well as test management, using OATS functional testing can be automated, load testing can be done with minimal effort and requirements and defect management can be achieved with minimal effort.

Use this link for a recorded overview presentation of Oracle Application Testing suite.



Oracle Test Manager

The task of defining test plans, acceptance criteria, testing deliverables and processes for any software development effort can face many different and evolving challenges, from identifying applicable processes to maintaining those decisions over time. Using a tool to centralize that information is highly recommended, Oracle offers OTM (Oracle Test Management) as a centralized solution for your test requirements, deliverables and issue tracking.

Open Script

OpenScript functional module is used to create functional tests scripts, using the functional module OpenScripts, scripts will replay and validate user interactions with the application trough the browser, by clicking on links, buttons and validating content as well as application functionality. OpenScript is the next generation scripting platform for Oracle Application Testing Suite, which can create automated test scripts for load testing and functional testing.

Combines an intuitive graphical scripting interface with a powerful Java IDE for creating test scripts providing much greater scripting flexibility while maintaining ease-of-use

Oracle Load Tester

Oracle Load Testing allows you to easily test the performance and scalability of your Web applications and Web services. Oracle Load not only stresses your application

to simulate the impact of end-user workloads, but also enables rigorous validation. Its integrated scripting platform cuts scripting time in half, eliminating weeks from a project's testing schedule. Oracle Load Testing is a component of Oracle Application.

Testing Suite, the centerpiece of the Oracle Enterprise Manager solution for comprehensive testing of packaged, Web and service-oriented architecture–based applications.

The following table is a mapping between the ATS tools and the applications suites to which they can be applied.

		Application Quality Management						
Tier	Solution / Test Target	ATS Test Manager for Web Applications	ATS Load Testing for Web Applications	ATS Functional Testing for Web Applications	Real Application Testing	Data masking	Real User Experience Insight	Guidance Document
Network infrastructure								
	any						M	
Hardware								
	any							
Virtual machine								
	any							
Operating System								
	any							
Database								
	any				Y	Y		
Middleware								
	SOA		Y*	Y*			Y	
	Web based		Y	Y			Y	
	any							
Application								
	Siebel		Y*	Y*			Y*	
	Peoplesoft		Y	Y			Y*	
	Hyperion		Y	Y			Y	
	E-Bussiness Suite		Y	Y			Y*	
	JD Edwards		Y	Y			Y	
	Web based		Y	Y			Y	
	any		M	M			Y	
End User								
	any		Y	Y			Y	

KEY	Tool Applicability to Product Suites

Y	Yes
Y*	Yes, accelerator available
M	Maybe/possibly under some circumstances

Back to Top

REFERENCES

For More Information

Contact HELPSAM_WW@ORACLE.COM for information about Oracle Testing and Quality Management Tools.

Back to Top

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.

TESTING AND QUALITY MANAGEMENT TOOLS PROCESS GUIDELINES

This document contains OUM supplemental process guidance for projects using Oracle® Application Testing Suite (OATS) Testing and Quality Management Tools.

TESTING (TE) PROCESS GUIDANCE

			Application Quality Management						
ID	Task	Work Product	ATS Test Manager for Web Applications	ATS Load Testing for Web Applications	ATS Functional Testing for Web Applications	Real Application Testing	Data masking	JDeveloper	Real User Experience Insight
Inception Phase									
TE.005.1	Determine Testing Requirements	Testing Requirements	Y						
Elaboration Phase									
TE.005.2	Determine Testing Requirements	Testing Requirements	Y						

TE.010	Develop Testing Strategy	Testing Strategy	Y						
TE.015.1	Develop Integration Test Plan	Integration Test Plan	Y						
TE.018.1	Prepare Test Data	Test Data		M	M		Y		
TE.020.1	Develop Unit Test Scripts	Unit Test Scripts	M		M			Y	
TE.025.1	Create System Test Scenarios	System Test Scenarios	Y		Y			M	
TE.030.1	Perform Unit Test	Unit-Tested Components	M		M	M		Y	
TE.035.1	Create Integration Test Scenarios	Integration Test Scenarios	Y	M	Y				
TE.038.1	Prepare Integration Test Environment	Integration Test Environment			Y		Y		
TE.040.1	Perform Integration Test	Integration-Tested Components	Y	M	Y				
TE.050.1	Develop System Test Plan	System Test Plan	Y						
TE.060.1	Prepare System Test Environment	System Test Environment					Y		
TE.070.1	Perform System Test	System-Tested Applications	Y		Y	Y			
TE.080	Develop Systems Integration Test Plan	Systems Integration Test Plan	Y						
TE.082	Develop Acceptance Test Plan	Acceptance Test Plan	Y						
Construction Phase									
TE.015.2	Develop Integration Test Plan	Integration Test Plan	Y						
TE.018.2	Prepare Test Data	Test Data		M	Y	M	Y		
TE.020.2	Develop Unit Test Scripts	Unit Test Scripts			M			Y	
TE.025.2	Create System Test Scenarios	System Test Scenarios			Y			Y	
TE.030.2	Perform Unit Test	Unit-Tested Components	Y		M	Y		Y	
TE.035.2	Create Integration Test Scenarios	Integration Test Scenarios		M	Y				
TE.038.2	Prepare Integration Test Environment	Integration Test Environment							
TE.040.2	Perform Integration Test	Integration-Tested Components	Y	M	Y				
TE.050.2	Develop System Test Plan	System Test Plan	Y						
TE.060.2	Prepare System Test Environment	System Test Environment					M		
TE.065	Perform Installation Test	Tested Installation Routines	Y		M				
TE.068	Configure System Test Environment	Configured System Test Environment							
TE.070.2	Perform System Test	System-Tested Applications	Y		Y	M			
TE.085	Prepare Systems Integration Test Environment	Systems Integration Test Environment					M		
TE.090	Develop Systems Integration Test Scenarios	Systems Integration Test Scenarios	Y	M	Y	M			
TE.100	Perform Systems Integration Test	Integration-Tested System	Y	M	Y	M			
Transition Phase									
TE.105	Prepare Users for Testing	Prepared Users for Testing	Y						
TE.110	Prepare Acceptance Test Environment	Acceptance Test Environment					M		
TE.120	Support Acceptance Test	Acceptance Test Results	Y						

KEY	Tool Applicability to TE Tasks
Y	Yes
M	Maybe/possibly under some circumstances

Back to Top

PERFORMANCE MANAGEMENT (PT) PROCESS GUIDANCE

			Application Quality Management						
ID	Task	Work Product	ATS Test Manager for Web Applications	ATS Load Testing for Web Applications	ATS Functional Testing for Web Applications	Real Application Testing	Data masking	JDeveloper	Real User Experience Insight
Inception Phase									
PT.010	Conduct Performance Management Workshop	Performance Management Workshop Results	Y						
Elaboration Phase									
PT.020	Define Performance Management Requirements and Strategy	Performance Management Requirements and Strategy	Y						M
PT.030	Define Performance Testing Strategy	Performance Testing Strategy	Y						
PT.040	Identify Performance Testing Models and Scenarios	Performance Testing Models and Scenarios	Y						
PT.050	Design Performance Test Scripts and Programs	Performance Test Scripts and Programs Design	Y	Y		Y			
PT.060	Design Performance Test Data and Load Programs	Performance Test Data and Load Programs Design		Y		Y	M		
Construction Phase									
PT.070	Build Performance Test Scripts and Programs	Performance Test Scripts and Programs	Y	Y					
PT.080	Construct Performance Test Environment and Database	Performance Test Environment and Database	Y	Y		Y	Y		
PT.090	Conduct Performance Test Dress Rehearsal	Validated Performance Test Environment and Environment	Y	Y					
Transition Phase									
PT.100	Execute Performance Test	Performance Test Results	Y	Y		Y			
PT.110	Create Performance Test Report	Performance Test Report	Y	Y		Y			
Production Phase									
PT.120	Conduct Production Performance Management	Managed Production Environment							Y

KEY	Tool Applicability to PT Tasks

Y	Yes
M	Maybe/possibly under some circumstances

Back to Top

ORACLE

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.

TESTING AND QUALITY MANAGEMENT TOOLS SUPPLEMENTAL TASK GUIDELINES

This document contains OUM supplemental task guidance for Testing and Quality Management Tools.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview.

Oracle Testing and Quality Management Tools - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Testing	
TE.025.1 Create System Test Scenarios	TE.025.1 Create System Test Scenarios
TE.020.2 Develop Unit Test Scripts	TE.020.2 Develop Unit Test Scripts
TE.025.2 Create System Test Scenarios	TE.025.2 Create System Test Scenarios
Performance Management	
PT.050 Design Performance Test Scripts and Programs	PT.050 Design Performance Test Scripts and Programs
PT.070 Build Performance Test Scripts and Programs	PT.070 Build Performance Test Scripts and Programs
PT.080 Conduct Performance Test Dress Rehearsal	PT.080 Conduct Performance Test Dress Rehearsal

Back to Top

TESTING (TE) GUIDANCE

TE.025.1 - Create System Test Scenarios

The task of defining test plans, acceptance criteria, testing deliverables and processes for any software development effort may present many different and evolving challenges, from identifying applicable processes to maintaining those decisions over time. Using a tool to centralize that information is highly recommended. Oracle offers OTM (Oracle Test Management) as a centralized solution for your test requirements, deliverables and issue tracking.

By coordinating the testing process in a single, unified platform, Oracle® Test Manager for Web Applications provides a comprehensive way to manage quality as a process throughout the Web application development lifecycle. Oracle Test Manager for Web Applications is tightly integrated with Oracle Functional Testing for Web Applications and Oracle Load Testing for Web Applications, maximizing the return on investment for your testing tools. It supports a process that consistently delivers high-quality, high-performance applications.

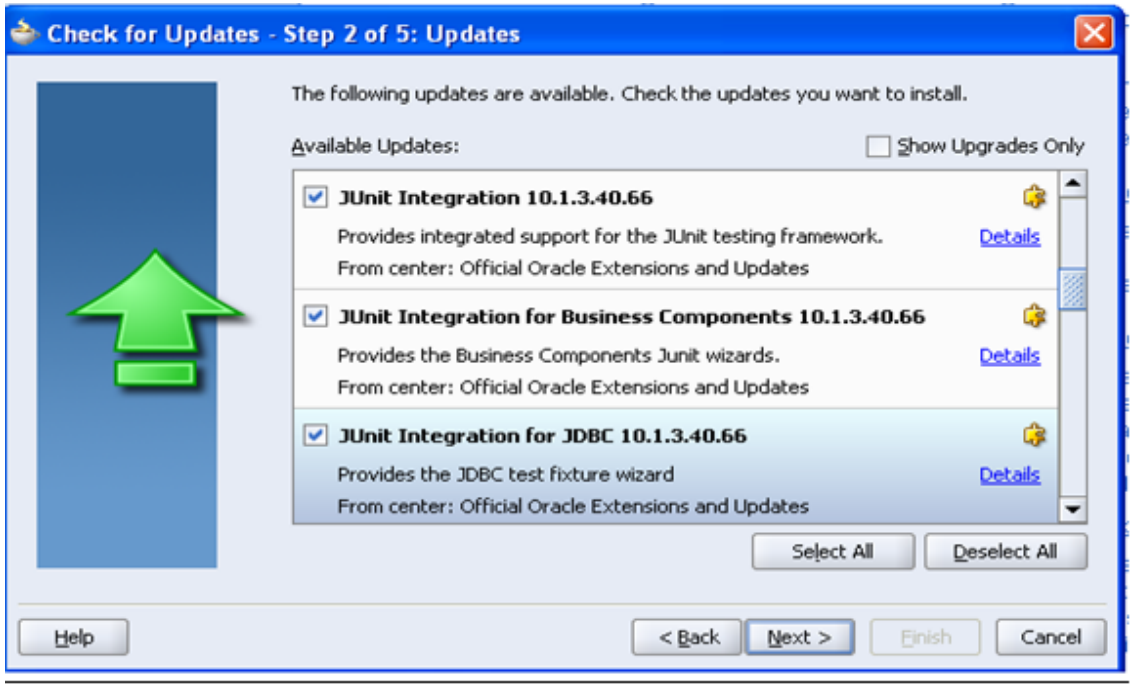
- The Requirements module allows testers to document and manage applications and test requirements before the testing process begins.
- The Tests module enables users to create detailed test plans, including automated and manual test cases.
- The Issues module allows testers and developers to capture and share issues found during the testing process, and then enables them to view issue status and gauge application readiness.

OTM can be used as a tracking tool and can also be used to execute automated and manual tests. OTM also has the option to schedule automated tests which makes this repetitive task more reliable.

TE.020.2 - Develop Unit Test Scripts

JDeveloper – JUnit

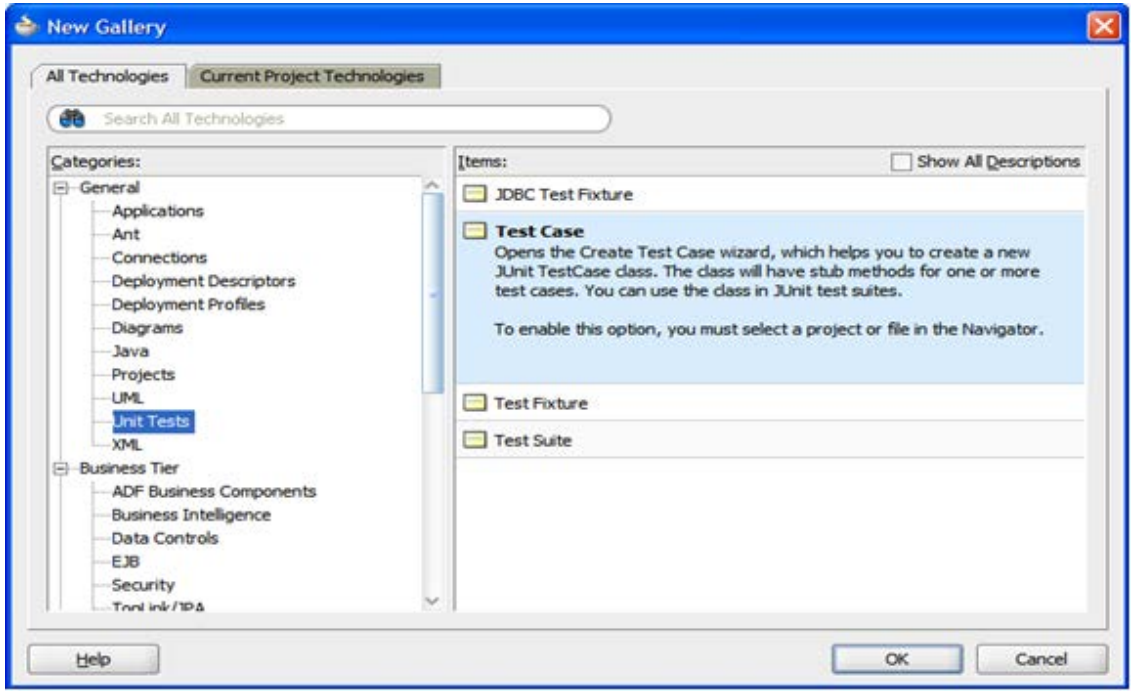
The de facto unit-testing framework for Java is JUnit (open source). You can plug JUnit easily into Oracle® JDeveloper from the menu: Help -> Check for Updates -> check the JUnit Extension -> next.



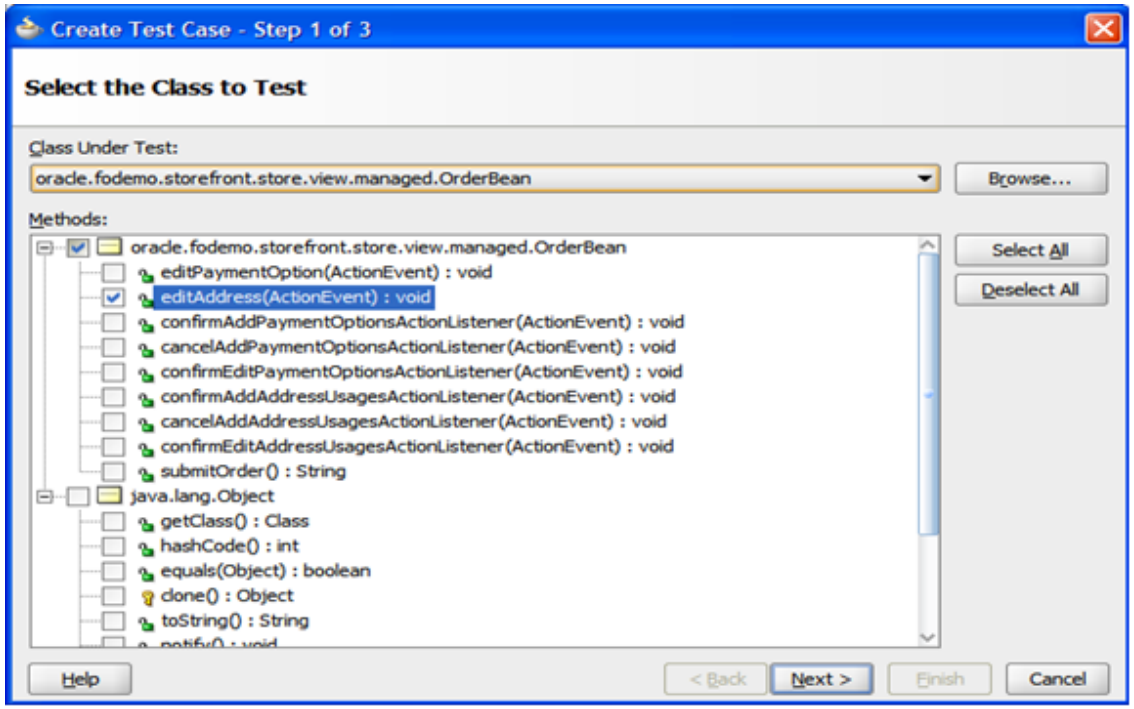
This will automatically download the JUnit plug-in files to your JDeveloper directory. JUnit can be used together with JDeveloper to test Java classes that run in a client JVM (Java Virtual Machine). Using the plug-in you can easily create unit test stubs for existing methods and create test suites for a set of existing unit tests. You can also create a set of reusable test fixtures.

Creating a A JUnit Test Case

Select the project or the particular case to be tested, -> New to Open the New Gallery, in the categories tree, expand the General tab and select Unit Tests -> Test Case



Select the method to be tested form the wizard:



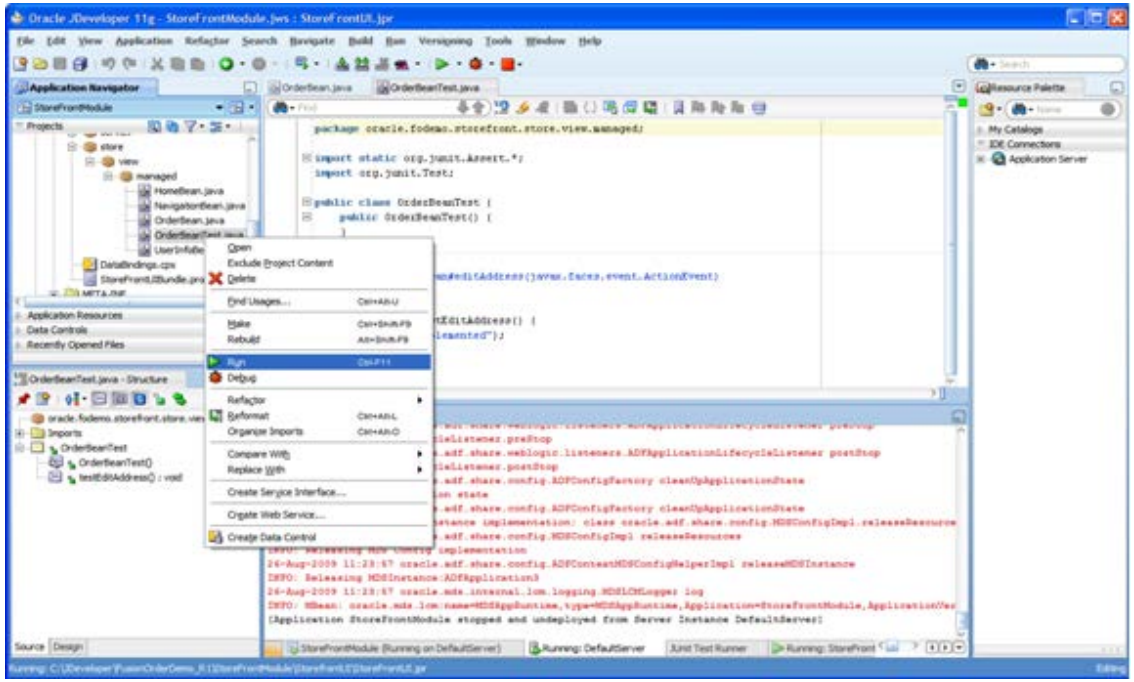
Click next then the class and Finish. Once that is done the new class will be added to the application navigator.

Open the new test case (The Java Class that has just been created) and include the values that check the function of the code.

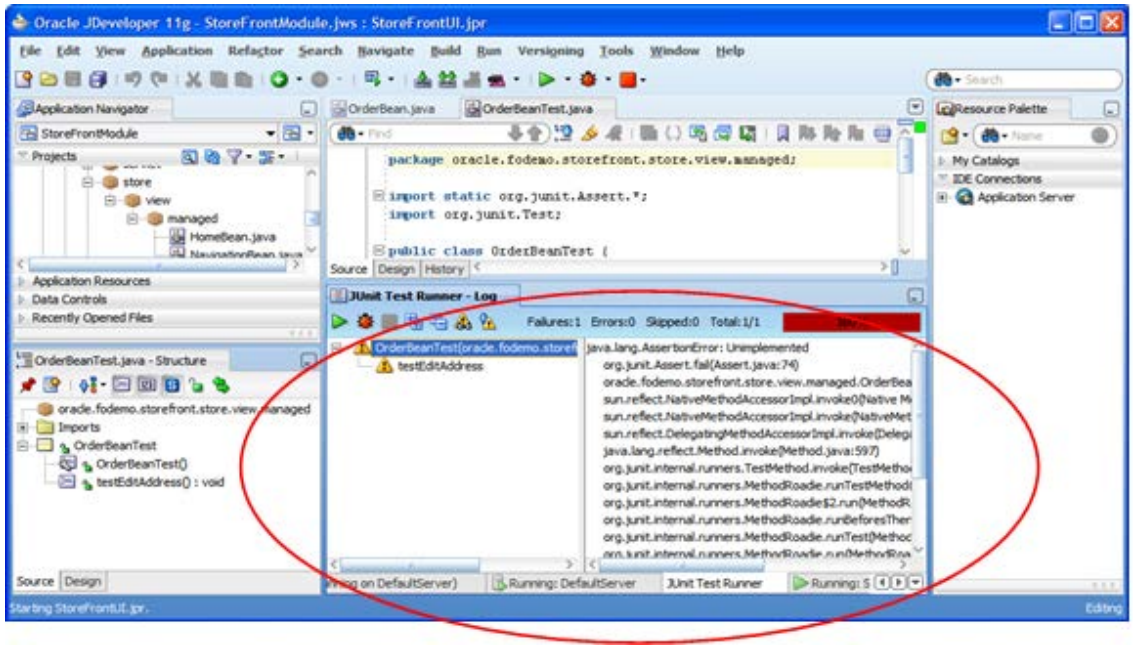
```
package oracle.fodemo.storefront.store.view.managed;

import static org.junit.Assert.*;
import org.junit.Test;

public class OrderBeanTest {
    public OrderBeanTest() {
    }

    /**
     * @see OrderBean#editAddress(javax.faces.event.ActionEvent)
     */
    @Test
    public void testEditAddress() {
        fail("Unimplemented");
    }
}
```

Running the Test Cases

After running the program and checking that it does not display any errors, we can run the JUnit tests. Right click on the test case and click on Run:



JDeveloper will show the progress of the tests in the results window:



Creating a Custom Test Fixture

Test Fixtures are used to create a fixed estate of a set of objects, the purpose of a test Fixture it to create a known environment where tests and results can be repeated.

Examples are:

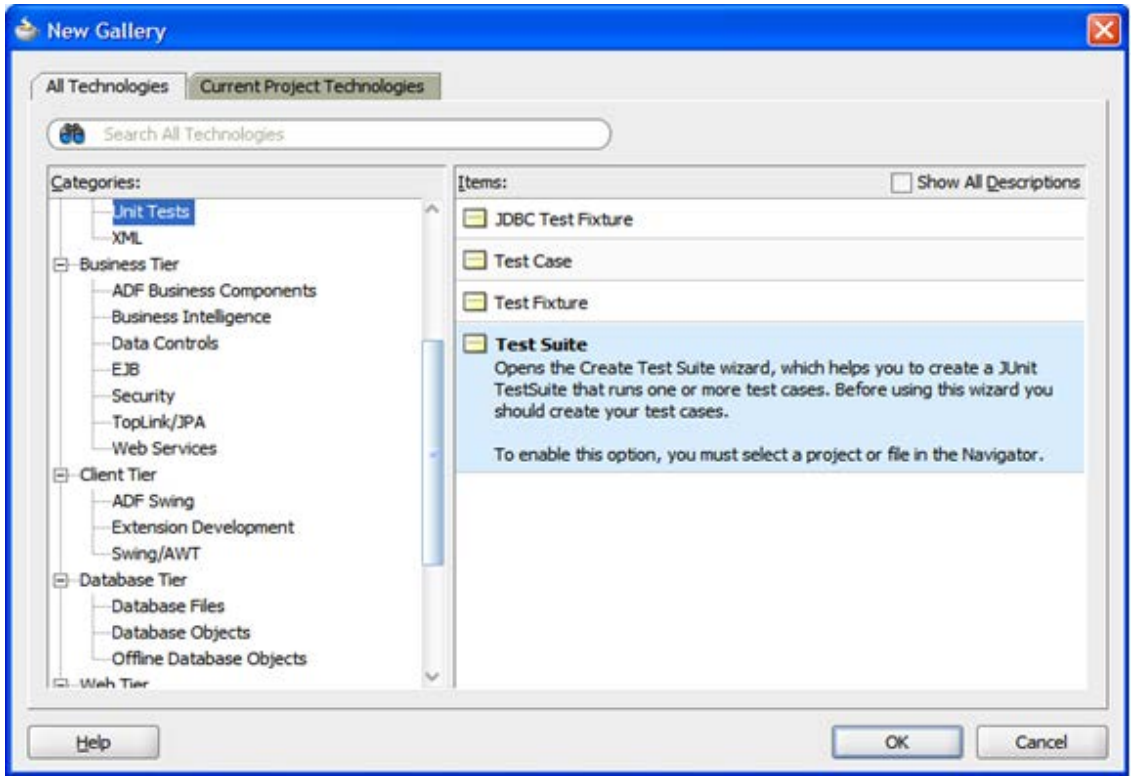
- loading a database with a set of data
- copying a known set of files
- setting up mock objects

Select the project or the particular case to be tested, -> New to Open the New Gallery, in the categories tree, expand the General tab and select Unit Tests -> Test Fixture, name the test and click OK.

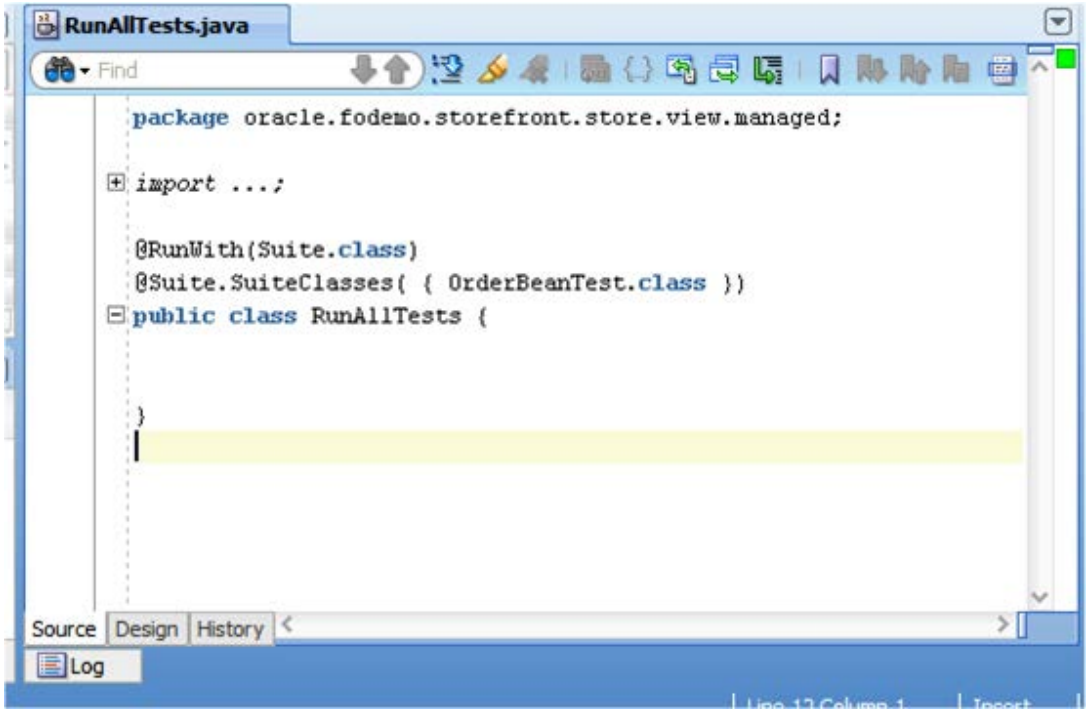
Running Multiple Test Cases

When multiple tests need to be executed a Test Suite needs to be created, The JUnit Test Wizard has options to insert a main() method and call the TestRunner class which will open the test TestRunner log and display the test results.

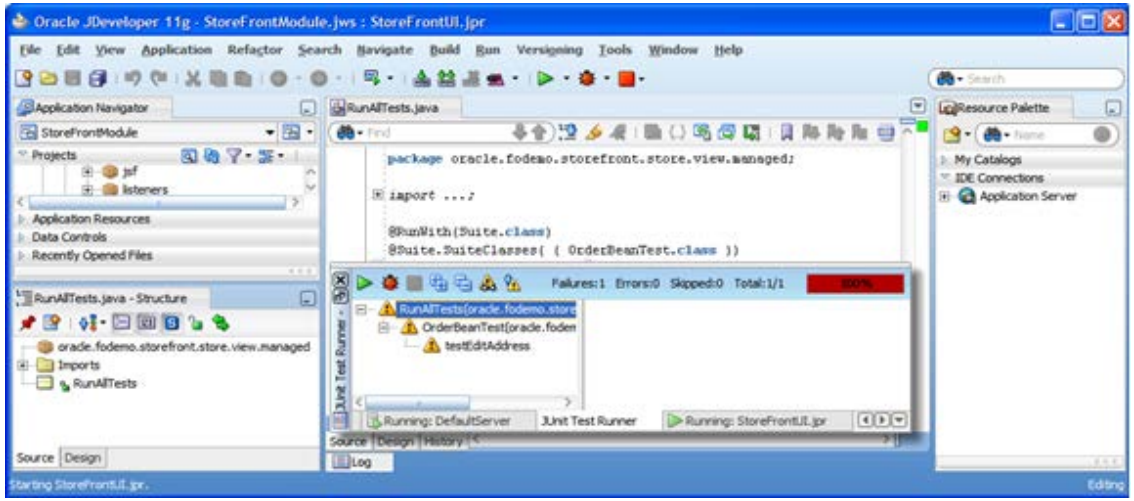
Select the project or the particular case to be tested, -> New to Open the New Gallery, in the categories tree, expand the General tab and select Unit Tests -> Test Suite name the test and click OK.



Name the test set and select the classes that will be used during the test.



Right click the class and select Run to run the test, the results will displayed in the results pane of JDeveloper.



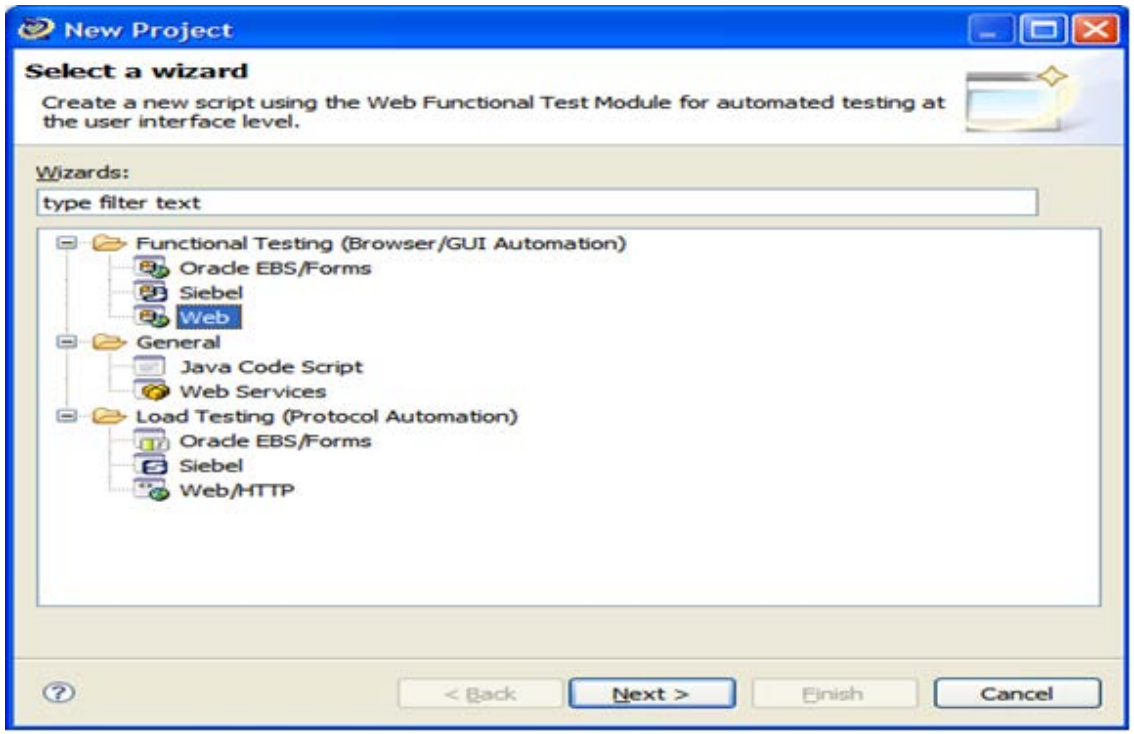
TE.025.2 - Create System Test Scenarios

OpenScript functional module is used to create functional tests scripts, using the functional module OpenScripts scripts will replay and validate user interactions with the application trough the browser, by clicking on links, buttons and validating content as well as application functionality.

OpenScript is the next generation scripting platform for Oracle Application Testing Suite, which can create automated test scripts for load testing and functional testing.

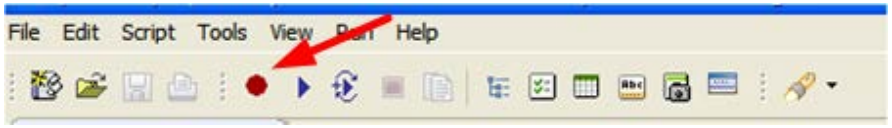
Combines an intuitive graphical scripting interface with a powerful Java IDE (Integrated Development Environment) to create test scripts, providing much greater scripting flexibility while maintaining ease-of-use.

To create a functional test script select: File -> New and Web under Functional Tests

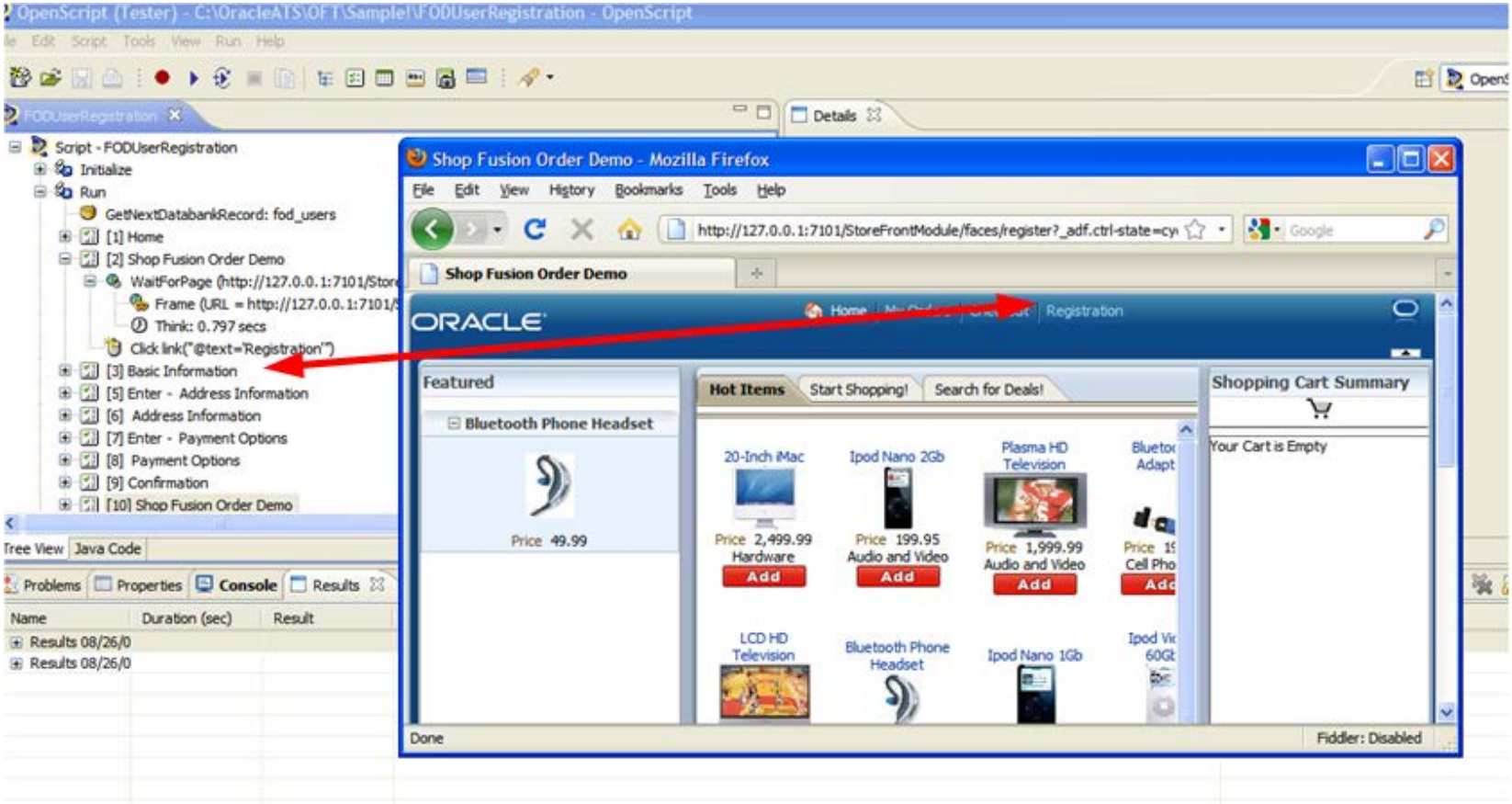


Select workspace and name the script.

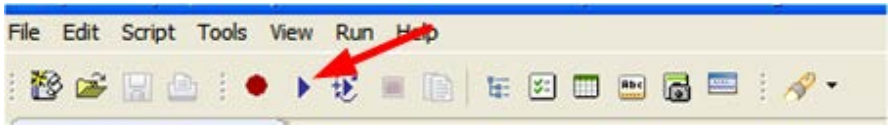
Start recording by clicking the record button.



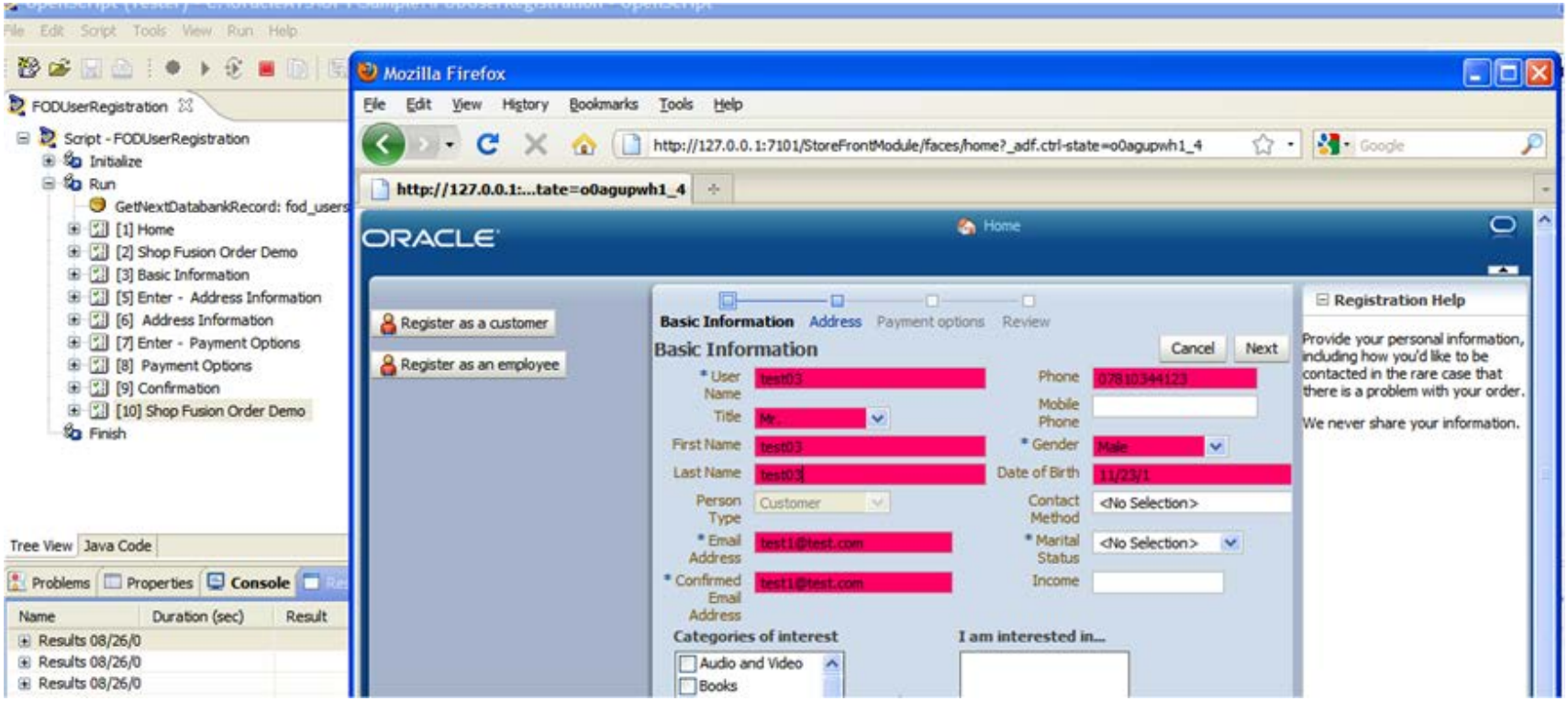
A browser will open, while interacting with the browser OpenScript will record the actions that are being performed.



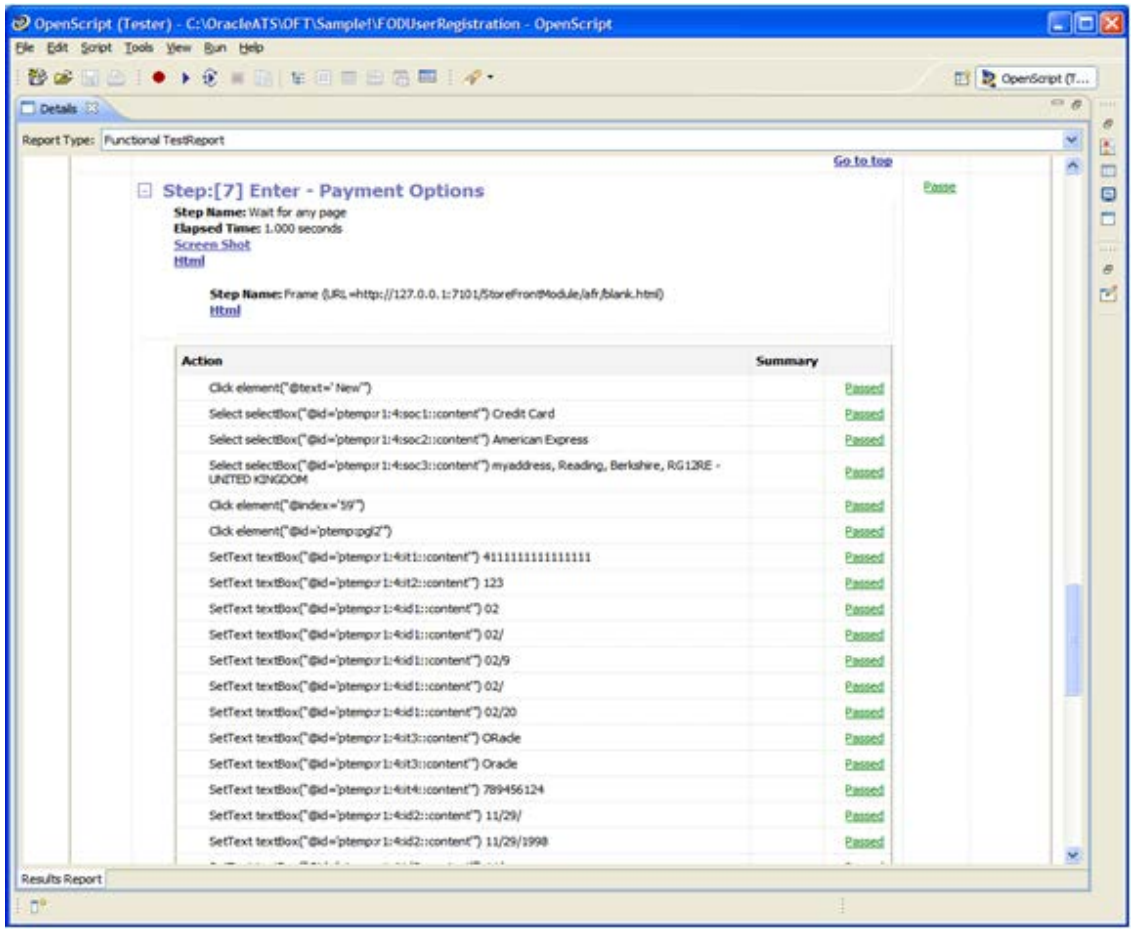
Once the script has been recorded, you will need to play it back. Click the playback button to execute the script.



Again the browser of your choice will open and the steps that you executed during recording will be reproduced.



Once the steps have been reproduced a report will be generated to show the results.



Back to Top

PERFORMANCE MANAGEMENT (PT) PROCESS GUIDANCE

PT.050 Design Performance Test Scripts and Programs
PT.070 Build Performance Test Scripts and Programs
PT.080 Construct Performance Test Environment and Database

ORACLE LOAD TESTING

Oracle® Load Testing allows you to easily test the performance and scalability of your Web applications and Web services. Oracle Load not only stresses your application to simulate the impact of end-user workloads, but also enables rigorous validation. The integrated scripting platform cuts scripting time in half, eliminating weeks from a project's testing schedule. Oracle Load Testing is a component of Oracle Application Testing Suite, the centerpiece of the Oracle Enterprise Manager solution for comprehensive testing of packaged, Web and service-oriented architecture–based applications.

It is composed of two components:

- OpenScript - Used to record, edit and verify scripts.
- OLT (Oracle Load Testing) - Used to reproduce multiple user interactions against the application.

OpenScript HTTP Module

OpenScript HTTP module is used to record the HTTP traffic that a user is generating while interacting with a browser or another application that generates HTTP calls. Its function is to record that traffic and interpret it, so it can be reproduced exactly as if it was being done by a user. You will then use OpenScript to validate and add validations to the script.

OpenScript Basic Module

OpenScript can also be used to test other non HTTP protocols, such as an Oracle Database, SSH, and Telnet.

To do so OpenScript offers a basic module that can be expanded using Java, the code can be integrated with OpenScript methods to validate and report errors.

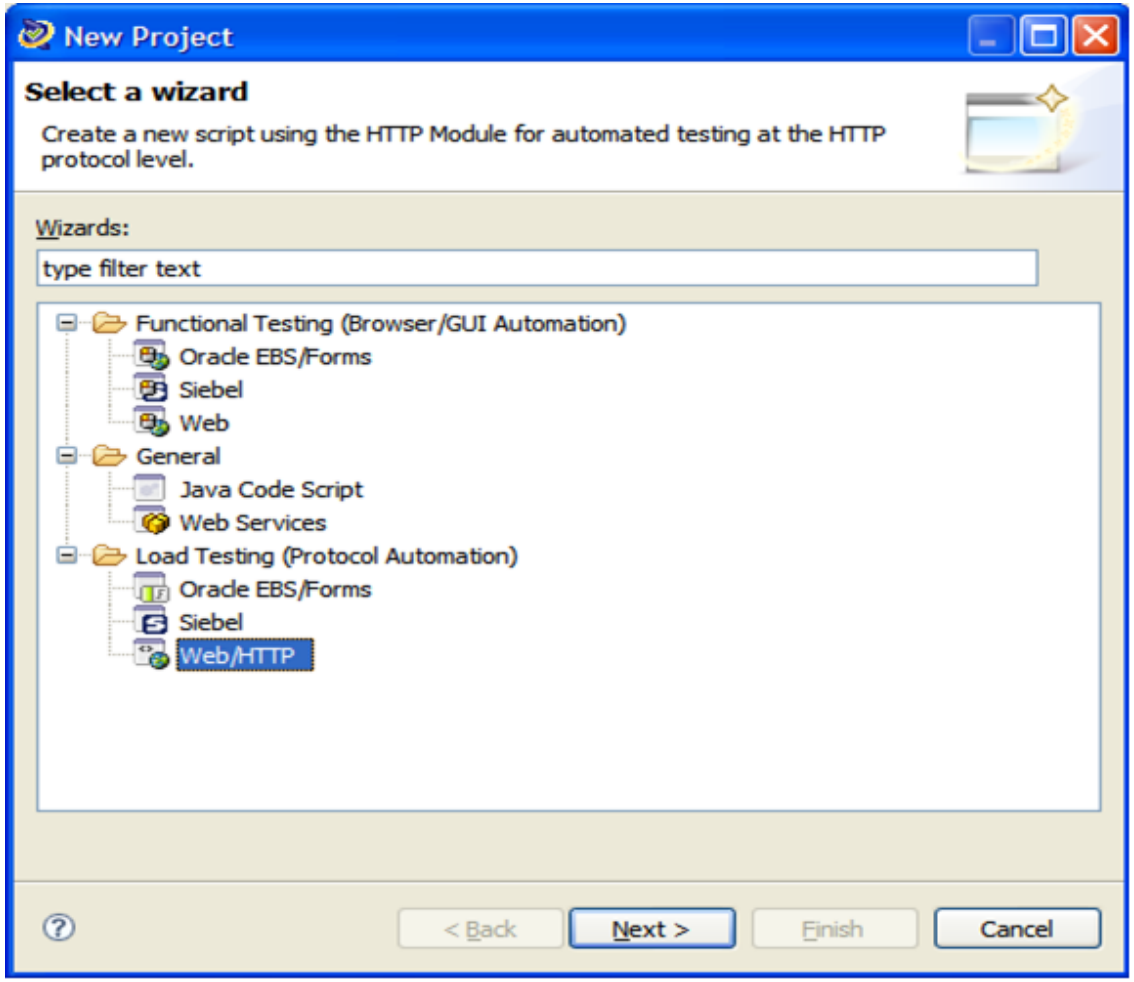
This script can later be called from OLT (Oracle Load Test) to generate load.

OpenScript WebServices Module

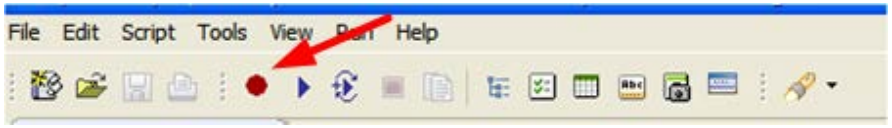
OpenScript can also be used to create scripts for WebServices, it does so by reproducing SOAP (Simple Object Access Protocol) calls. The scripts can then be edited, parameterized and validated.

CREATING HTTP LOAD SCRIPTS

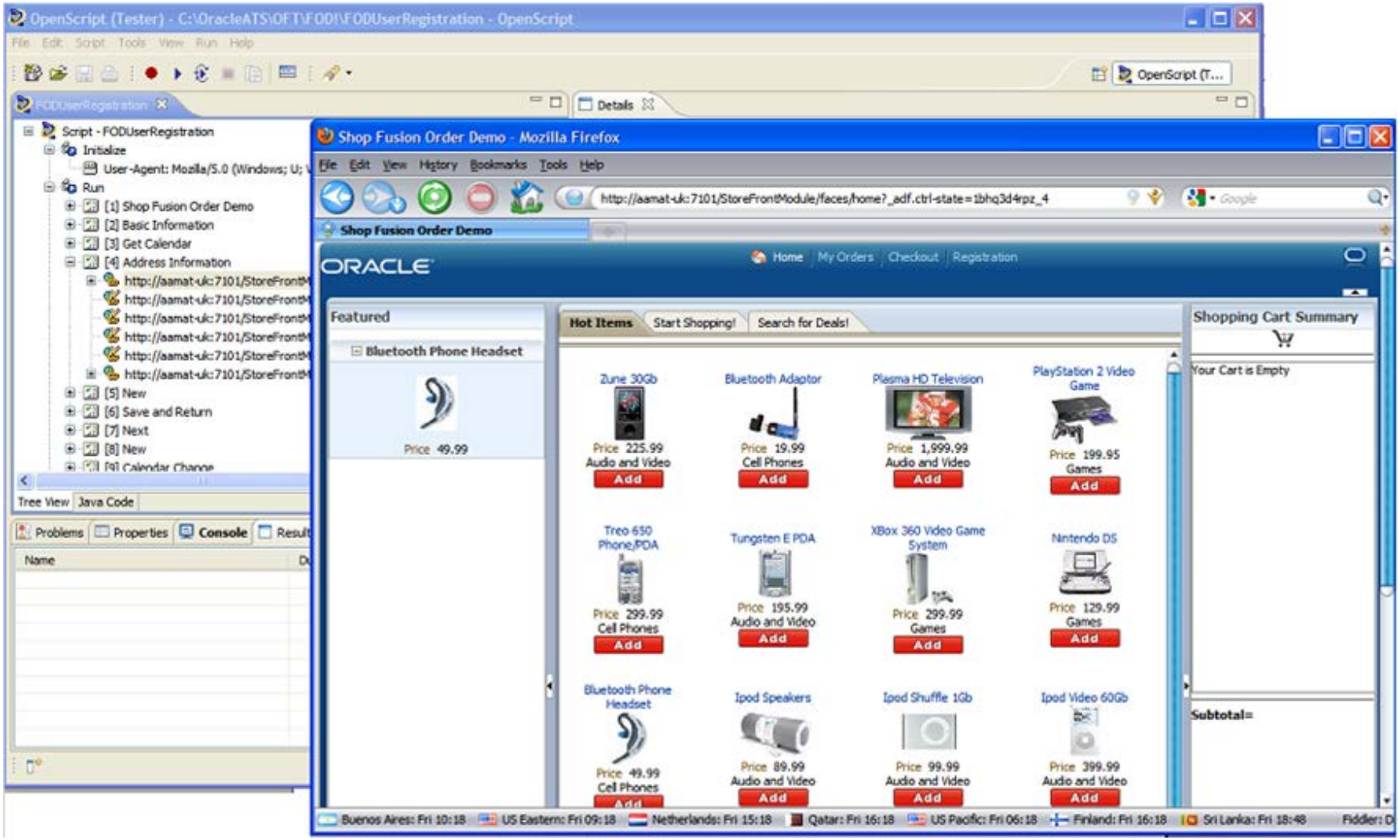
We will use OpenScript to create these scripts; first we will select the Web/HTTP module to generate the script.



Select next, enter a name and select the workspace where you want to store it. You will start recoding by clicking on the record button:

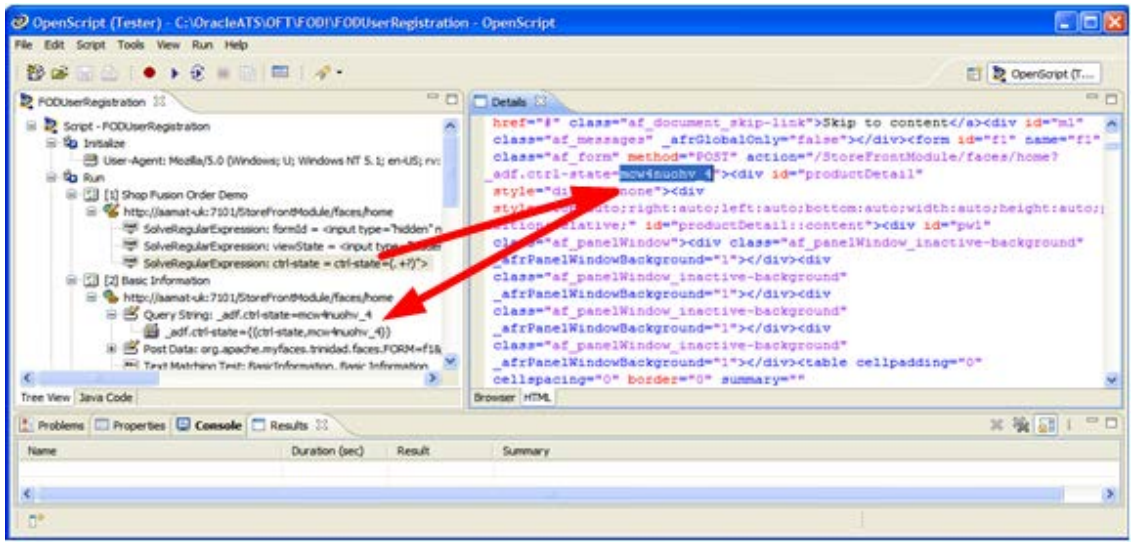


After clicking on the icon your preferred browser will open, interact with the browser to record the HTTP transactions.



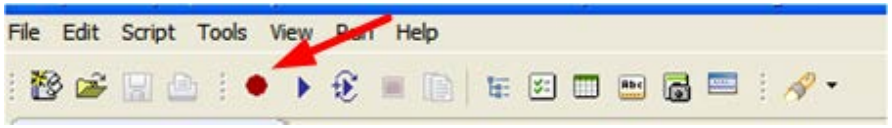
Validating HTTP Scripts

Use OpenScript to correlate and validate your scripts.

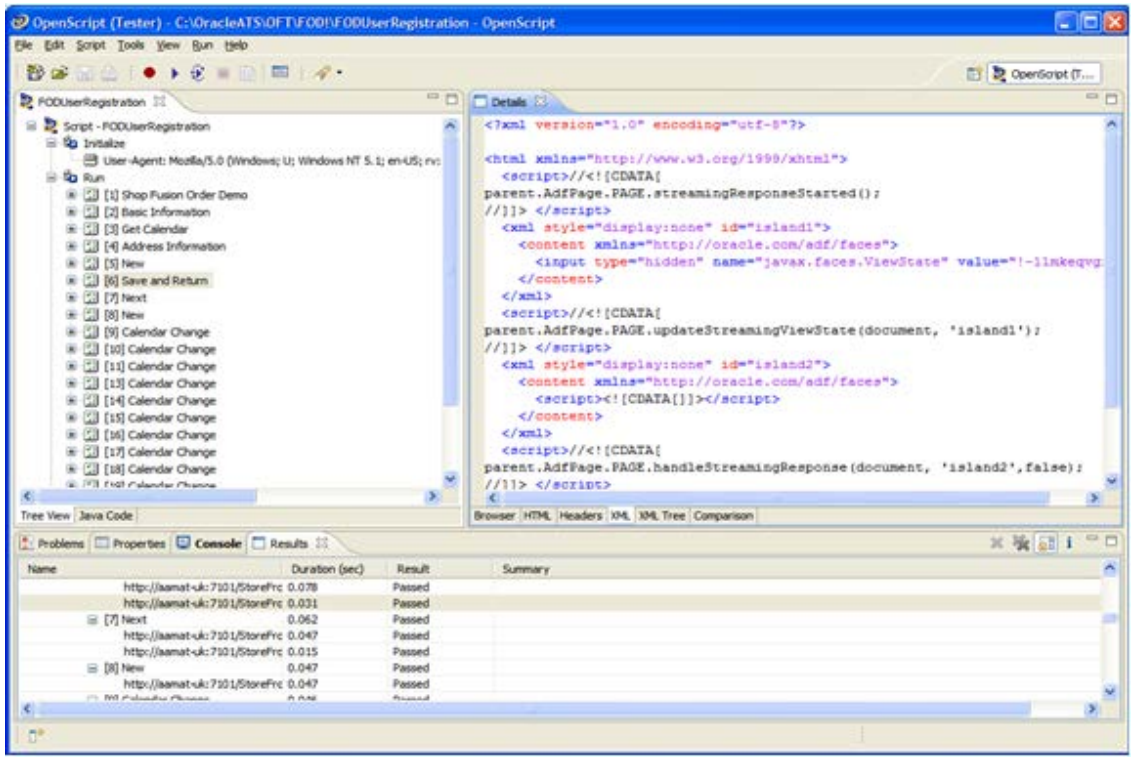


The above picture shows a parameter being substituted using correlation.

To play back the scripts in OpenScript use the Playback button.

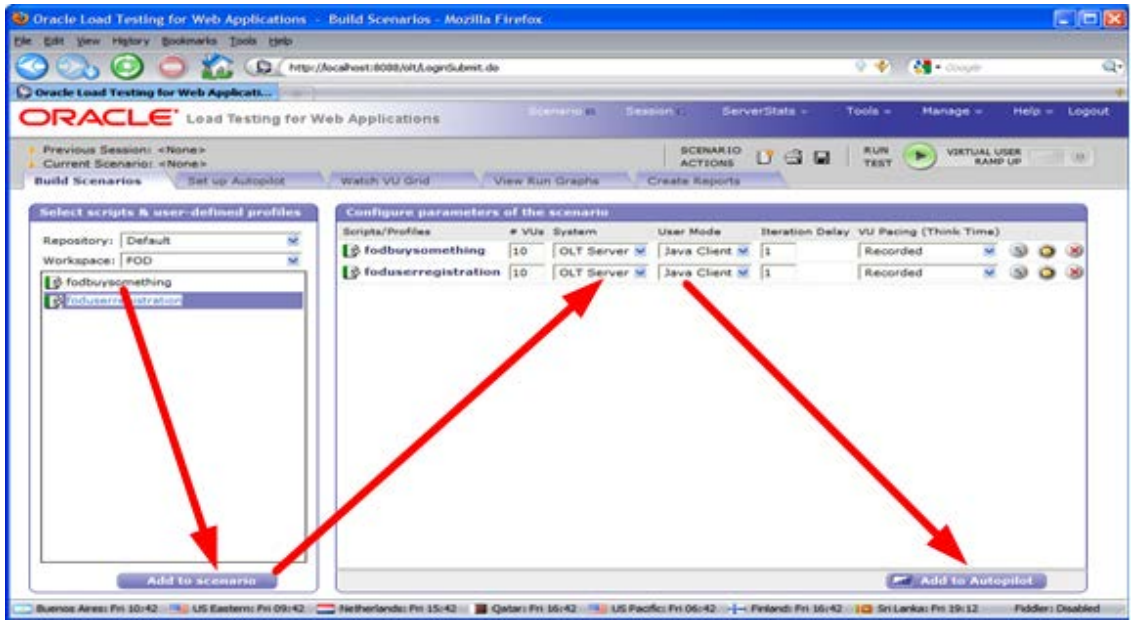


As the script plays back, the Details pane will display the contents received by the server. It will also report any HTTP errors and perform any user defined validations.



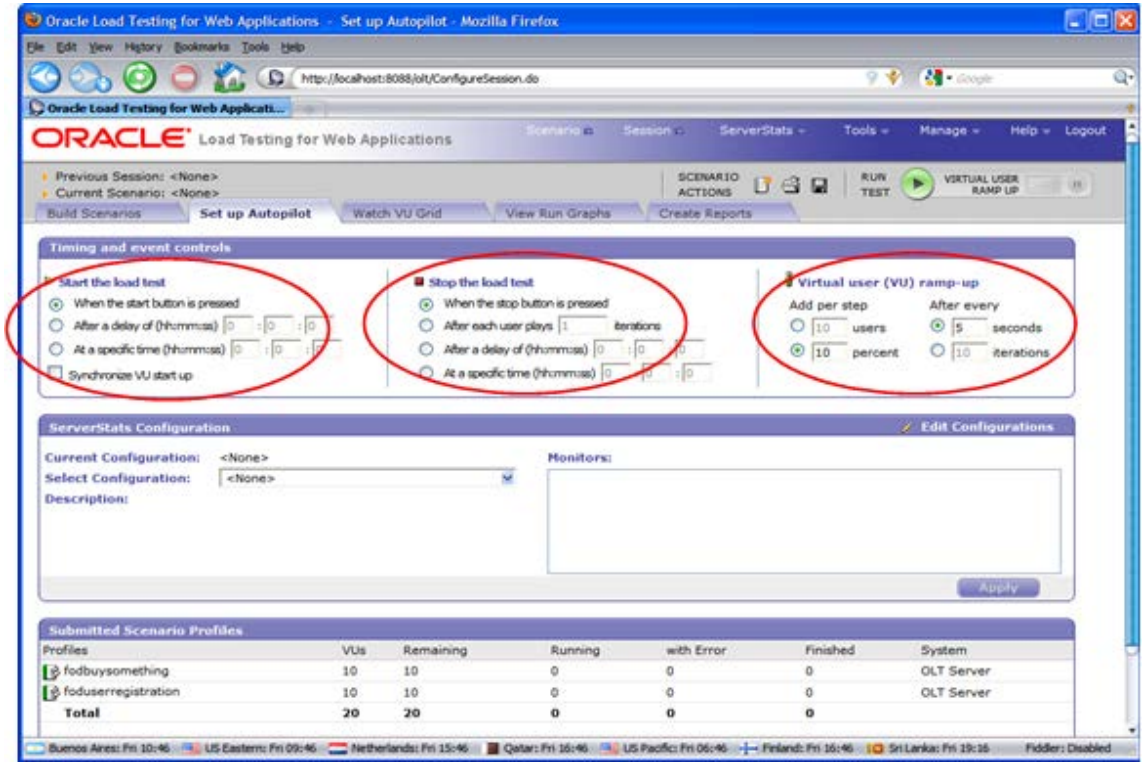
Once scripts have been validated and verified that it is time to start generating load for the performance test, you will do so using OLT.

First you will select the desired scripts from OLT and build the scenario. Once you have defined the scenario you will add them to the autopilot.

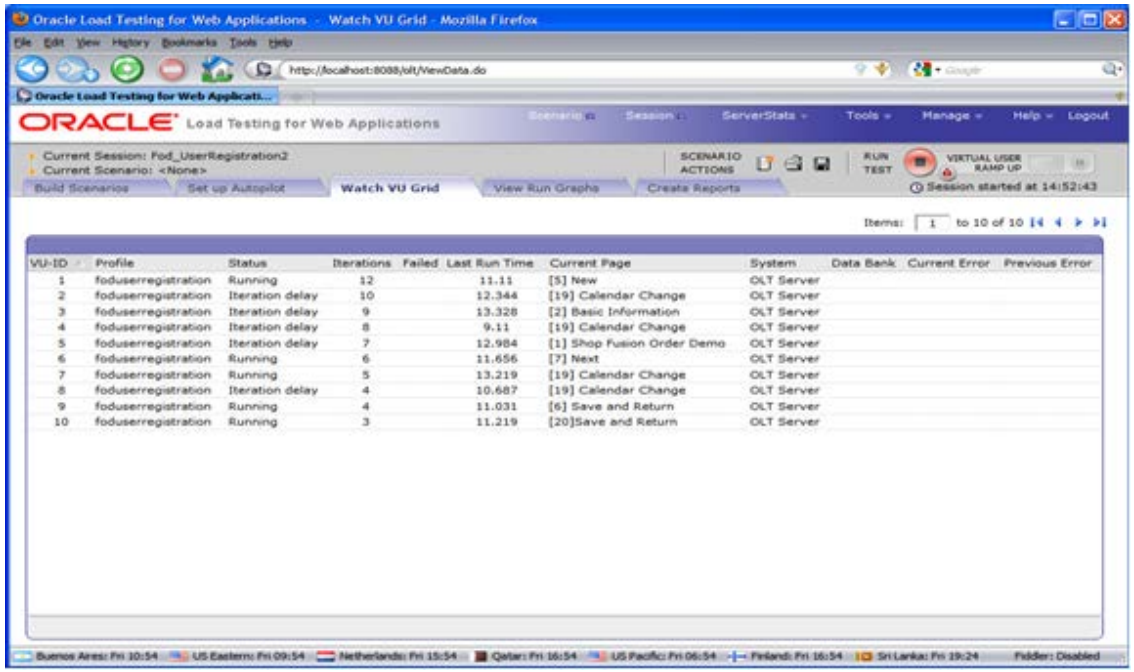


Once you have added the scenario to the autopilot, you can define how you would like to start and end the test, and user ramp up. You can also indicate if you would like

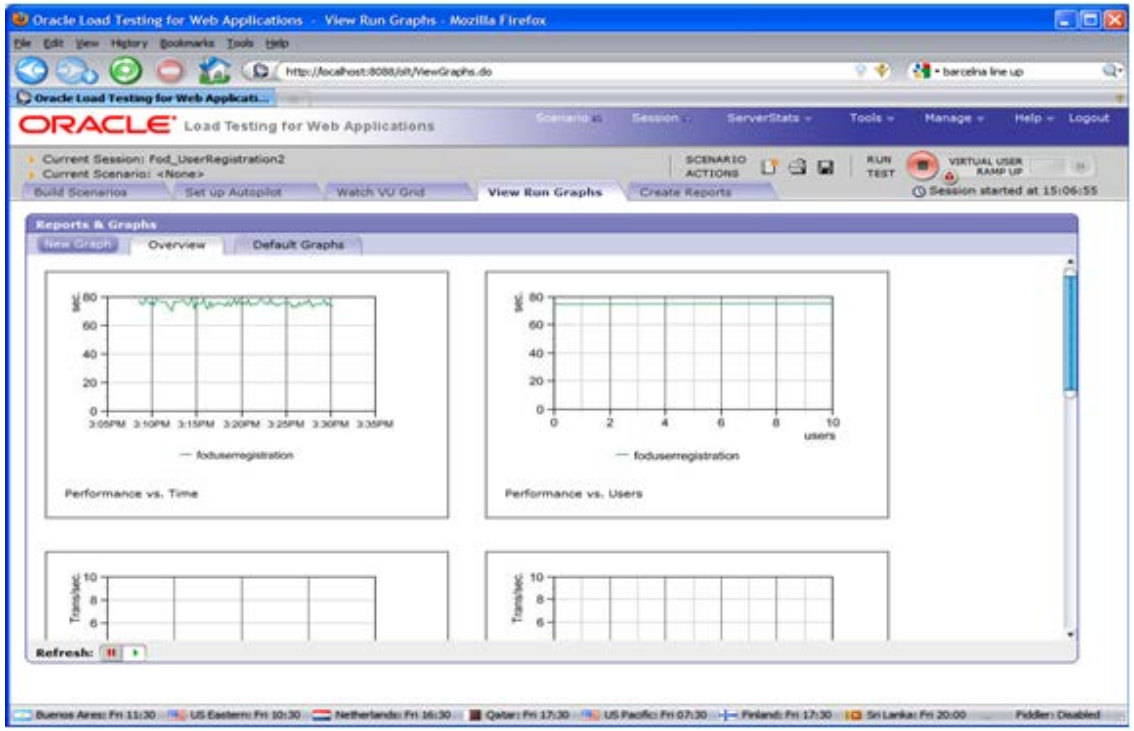
to add any system metrics.



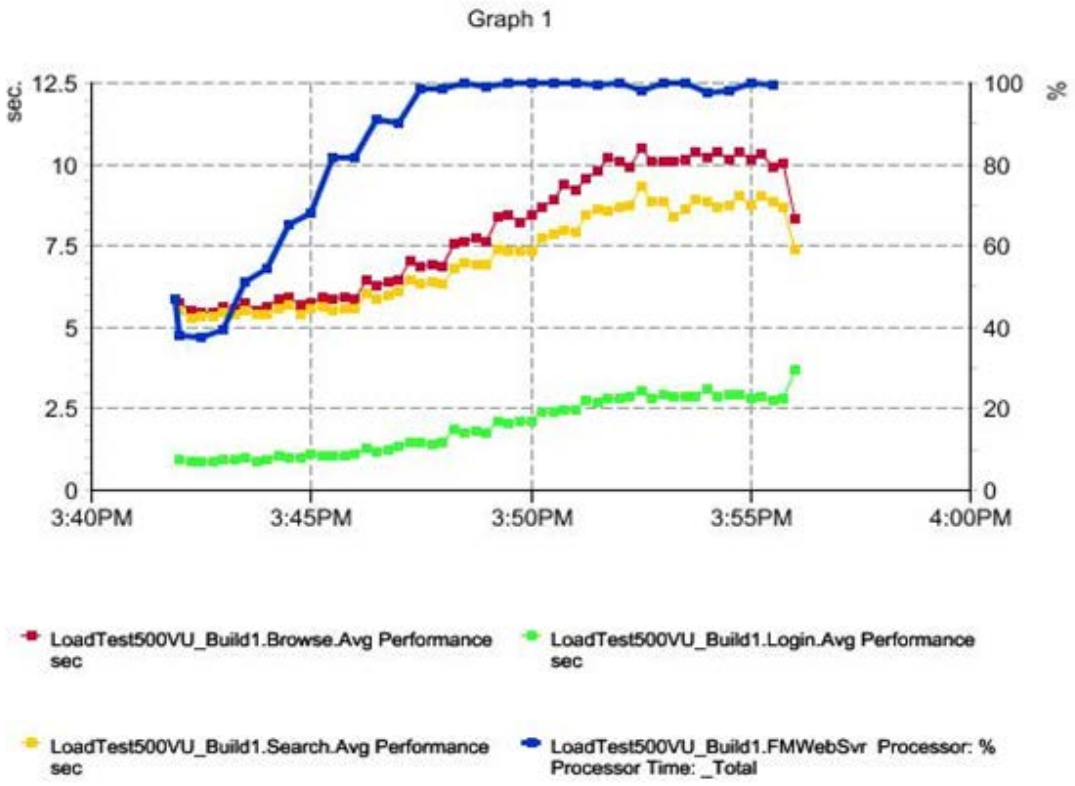
Click on the play button and name the session to start the test. As the test is running, you can see the VU (Virtual User) grid being updated.



The Run graphs can also be used to show the status of the application.



Finally use the reports tabs to view any metrics and build custom reports.



Back to Top

ORACLE

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.